

Known as both a philosopher and a scientist, Aristotle laid the foundations for scientific methodology by introducing the concept of logic. Using reasoning to support explanations about natural phenomena, his writings spanned almost every field of science. His ideas held sway for centuries and were profoundly influential on both Western and Arabic thinking.

MILESTONES

PLATO'S PUPIL

Joins Plato's Academy in 367 BCE, and remains there until his mentor's death 20 years later.

BIOLOGICAL STUDIES

In 348 BCE, begins to write *On the Parts of Animals* and *The Generation of Animals*, naming 500 species.

ROYAL CHARGE

Becomes tutor to 13-year-old Prince Alexander of Macedon, later Alexander the Great, in 342 BCE.

PROLIFIC WORKS

Founds the Lyceum in 335 BCE, and begins researching and writing many of his treatises.

Originally from Macedonia, northeastern Greece, Aristotle traveled to Athens at the age of 17 to study at the Academy of Plato, the center of learning in the Greek world. When Aristotle left, he used what he had learned there to develop an approach that was based more on observation than on theory. For Aristotle, facts about the natural world were as interesting as the process of reasoning; in this way, he was both one of the first scientists and a philosopher.

Aristotle realized that rational thinking was fundamental to science, and he developed an ordered approach to logical reasoning. Moreover, he understood that science requires practical data, which is gained by observation. Aristotle saw that such information can form the premises (starting points) from which conclusions can be derived about the world by means of logic.

Aristotle's work was extraordinarily wide-ranging in its subject matter. He wrote some 200 treatises, which developed the framework for the fields of physics, chemistry, biology, astronomy, logic, and mathematics. One of his most important surviving written works, *Physics*, remained unchallenged until the Middle Ages. He believed that the science of physics encompassed almost everything there is

Aristotle's Lyceum was known as the "Peripatetic School," after the Greek *peripatoi*, meaning "walking," because the teaching was conducted while strolling around.



“For nature is everywhere **the cause of order.**”

Aristotle